



MA 653P: Computational Financial Modelling Lab

Course Code	MA 653P
Course Name	Computational Financial Modelling Lab
L-T-P-C	0-0-2-1
Prerequisites	MA 653 Computational Financial Modelling should be taken concurrently or earlier. Some programming knowledge in C/MATLAB/R/Spread sheets Packages.
Intended for	3rd and 4th Year B. Tech / PG
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	8th Senate (Oct 2015)

Course Outline

- **Module I**

Introduction to R statistical software. [2 Lectures]

- **Module II**

Classical Markowitz portfolio model, portfolio under higher order moments and fuzzy portfolio models in R. [3 Lectures]

- **Module III**

Financial Time series analysis, stochastic volatility models and non parametric time series and technical analysis using R. [5 Lectures]

Factor Models, Regression, classification and clustering analysis of financial data in R. [2 Lectures]

- **Module IV**

Text Books

1. Tsay, Ruey S., **Analysis of financial time series**, Vol. 543. John Wiley & Sons, 2005.
2. Wurtz, Diethelm, Yohan Chalabi, William Chen, and Andrew Ellis, **Portfolio optimization with R/Rmetrics**, Rmetrics, 2009.
3. Sheather, Simon, **A modern approach to regression with R**, Vol. 58. Springer, 2009.
4. Pfaff, Bernhard, **Financial risk modelling and portfolio optimization with R**, John Wiley & Sons, 2012.